

EDUCATION

- **The University of Edinburgh** Edinburgh, United Kingdom
PhD in Informatics: Robotics, Computer Vision, Computer Graphics and Animation *Sep 2019 – Oct 2023*
- **The University of Edinburgh** Edinburgh, United Kingdom
MSc in Artificial Intelligence *Sep 2018 – Aug 2019*
- **Ulsan National Institute of Science and Technology (UNIST)** Ulsan, South Korea
Bachelor in Electrical Engineering, Computer Science and Engineering *Mar 2014 – Feb 2018*
 - **Contribution Award**¹: Chairman's Award
 - **GPA of major subjects**: 4.00/4.3 (96.0/100)
 - **Total Grade Point Average**: 3.79/4.3 (93.9/100)

PUBLICATIONS

- K Kim, SN Gowda, P Eustratiadis, A Antoniou, RB Fisher, “Adversarial Augmentation Training Makes Action Recognition Models More Robust to Realistic Video Distribution Shifts”, *4th International Conference on Pattern Recognition and Artificial Intelligence, ICPRAI 2024*
- K Kim, D Moltisanti, O Mac Aodha, L Sevilla-Lara, “An Action Is Worth Multiple Words: Handling Ambiguity in Action Recognition”, *33rd British Machine Vision Conference, BMVC 2022*
- K Kim, SN Gowda, O Mac Aodha, L Sevilla-Lara, “Capturing Temporal Information in a Single Frame: Channel Sampling Strategies for Action Recognition”, *33rd British Machine Vision Conference, BMVC 2022*
- SN Gowda, L Sevilla-Lara, K Kim, F Keller, M Rohrbach, “A new split for evaluating true zero-shot action recognition”, *DAGM German Conference on Pattern Recognition, 2021*
- M. S. Ryoo, K. Kim and H. J. Yang, “Extreme Low Resolution Activity Recognition with Multi-Siamese Embedding Learning,” *AAAI Conference on Artificial Intelligence*, New Orleans, Louisiana, February 2018
- **Patent**: Falling out of hair management system
<https://patents.google.com/patent/KR20140094301A/en>

EXPERIENCE

- **Deargen Inc.** Seoul, South Korea
AI Platform Lead *Jan 2024 –*
 - **Full-Stack Dental EMR with LLM agents – Zero to Pilot Deployment**: Principal developer: Electron app with LLM agents operating on the entire clinical database.
 - **AI Drug Discovery – Molecular Glue / Targeted Protein Degradation Research**: 3D surface-based geometric deep learning to predict molecular glue ternary complexes (E3 ligase–molecular glue–substrate) for targeted protein degradation; built the full preprocessing, training, and virtual screening pipeline, narrowing ~1M compounds to ~100 for wet-lab testing.
 - Wrote Dijkstra algorithm in Rust for 23x speed improvement over Python networkx:
<https://github.com/deargen/rust-graph>
 - **Data Curation for DTI (Drug Target Interaction)**: <https://github.com/deargen/bio-data-to-db>
- **Innerverz Inc.** Seoul, South Korea
Image Generation AI Researcher & Developer *Oct 2023 – Jan 2024*
 - **Implement FP-LIIF (CVPR 2023 paper)**: Face segmentation technology using super-resolution
 - **Stable Diffusion Personalization**: Research and develop DreamBooth and LoRA. Made the model 20% faster than Huggingface by applying flash attention more efficiently (qkv-packed version). Launched an app called KastingOn. <https://www.aifnlife.co.kr/news/articleView.html?idxno=22729>
 - **SBS News De-Aging and Aging Model Demo**: Applied spatial and temporal denoising:
<https://youtu.be/jOW68s4bsT4>

¹Students who have shown especially good behaviour or enhanced the university's honour can be given a contribution award. The contribution award winner shall be selected by deliberation of the Committee based on recommendations from the head of the school (department) in each school (department). In 2018, among 704 undergraduates from various departments, 6 people have been awarded.

- EgoVid Inc.**

Ulsan, South Korea

May 2016 – Dec 2017

Machine Learning Researcher & Developer

 - Realtime Demonstration Running on Embedded Device at CVPR 2017:** Demonstrated my work running on NVIDIA Jetson TX2 and helped my colleague for implementing different demonstration about Video Anonymisation Algorithm at CVPR (Computer Vision Pattern Recognition) 2017 conference.
https://youtu.be/7jkSum_pj9o?t=25s
 - UC Berkeley Entrepreneurship Programme:** Visited UC Berkeley Sutardja Center of Entrepreneurship & Technology for 9 weeks to exchange ideas about making a start-up company.
UC Berkeley News Article: <http://scet.berkeley.edu/keeping-personal-machine-learning-meets-egovid/>
 - Attended Conferences:** AAAI Conference on Artificial Intelligence 2017, San Francisco, California, USA
Asilomar 2016, Pacific Grove, California, USA
Ubicomp 2016, Heidelberg, Germany
 - Linux ML Computing Server Setup and Maintenance:** Permissions, security, backup, scheduling, infrastructure, etc.
- GitHub Projects (Terminal-based IDE)**

Dec 2022 – Mar 2023

Maintainer

 - Nvim Treesitter Textobjects - 2.8k stars:** Select meaningful blocks of code using tree-sitter AST
 - Nvim Hand Gesture - 96 stars:** Webcam hand gesture recognition to control IDE
 - Jupynium - 775 stars:** Jupyter Notebook integration for NeoVim
 - Treemux - 205 stars:** File explorer in terminal (Tmux plugin)

AWARDS

- NAVER UNIST Undergraduate Poster Award (NAVER UUPA)**

NAVER Corp., UNIST

1st place: Poster presentation of Extreme Low Resolution Activity Recognition

Dec 2017
- Korea Supercomputing Challenge (KSC 2015)**

KISTI Supercomputing Education Center

5th place: MPI parallel computing competition

Oct 2015
- HeXATHON (Hackathon)**

UNIST, NAVER Corp.

1st place: QR code waiting system implemented with Raspberry Pi

May 2015
- Korea International Science and Engineering Fair (KISEF 2012)**

National Science Museum

Exhibited and Demonstrated PowerUpdater2 at Daejeon Convention Center.

Jan 2012
- Korea Olympiad in Informatics (KOI 2011)**

National Information Society Agency

Bronze medal: Demonstrated PowerUpdater2 (MFC, C++)

Sep 2011
- PhD Full Funding & Stipend**
- Bachelor's Degree Full Funding & Stipend**

TEACHING SUPPORT

- Marker of Introductory Applied Machine Learning**

The University of Edinburgh

Marking coursework submissions for the MSc module.

Oct 2020 – Jan 2021
- Demonstrator/marker of Advanced Vision**

The University of Edinburgh

Ran lab sessions for the Advanced Vision (MSc module).

Jan 2020 – May 2020
- Demonstrator of Machine Learning Practical**

The University of Edinburgh

Ran lab sessions for the Machine Learning Practical (MSc module).

Sep 2019 – Dec 2019
- Tutor of Object-Oriented Programming**

The University of Edinburgh

Ran 2 tutorial sessions of OOP Java programming tutorials with Eclipse IDE.

Jan 2019 – Apr 2019

 - Nominated for the **Best Student Who Tutors Award**
- Tutor of Imperative Programming**

The University of Edinburgh

Ran tutorial sessions of Java programming with Greenfoot IDE.

Nov 2018
- Tutor of Engineering Programming I**

UNIST

Ran tutorial sessions for 3 terms.

Sep 2014 – Dec 2015

SKILLS

- **Languages:** Python, TypeScript, C, C++, C#, Lua, Bash
- **Deep Learning:** PyTorch, Keras
- **Parallel Programming:** MPI, CUDA, PyTorch DDP (Distributed Data Parallel)
- **Tools:** Git, GitHub, Docker, Kubernetes
- **Research:** MATLAB, OpenCV
- **Embedded Computer:** NVIDIA Jetson TX2, Raspberry Pi (IoT)
- **Database & Web:** React, PHP, MySQL, PostgreSQL, FastAPI, Flash ActionScript, WordPress
- **Others:** NXT Robot C, Electron/WPF/MFC (desktop app)

LANGUAGES

- **English:** Advanced: IELTS 7.5
- **Korean:** Native
- **Spanish, French, Italian, Portuguese:** Intermediate